

Log in as root user and make a new user account. We now access user through command line.

### User details

**User name**

user-for-cli

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ \_ - (hyphen)

☒ **Provide user access to the AWS Management Console - optional**

If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

**Console password**

☐ **Autogenerated password**

You can view the password after you create the user.

☒ **Custom password**

Enter a custom password for the user.

\*\*\*\*\*

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & \* ( ) \_ + - (hyphen) = [ ] { } ' "

☐ Show password

☐ **Users must create a new password at next sign-in - Recommended**

Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

**!** If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

☐ **Add user to group**

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

### Permissions policies (1/1399)

Choose one or more policies to attach to your new user.

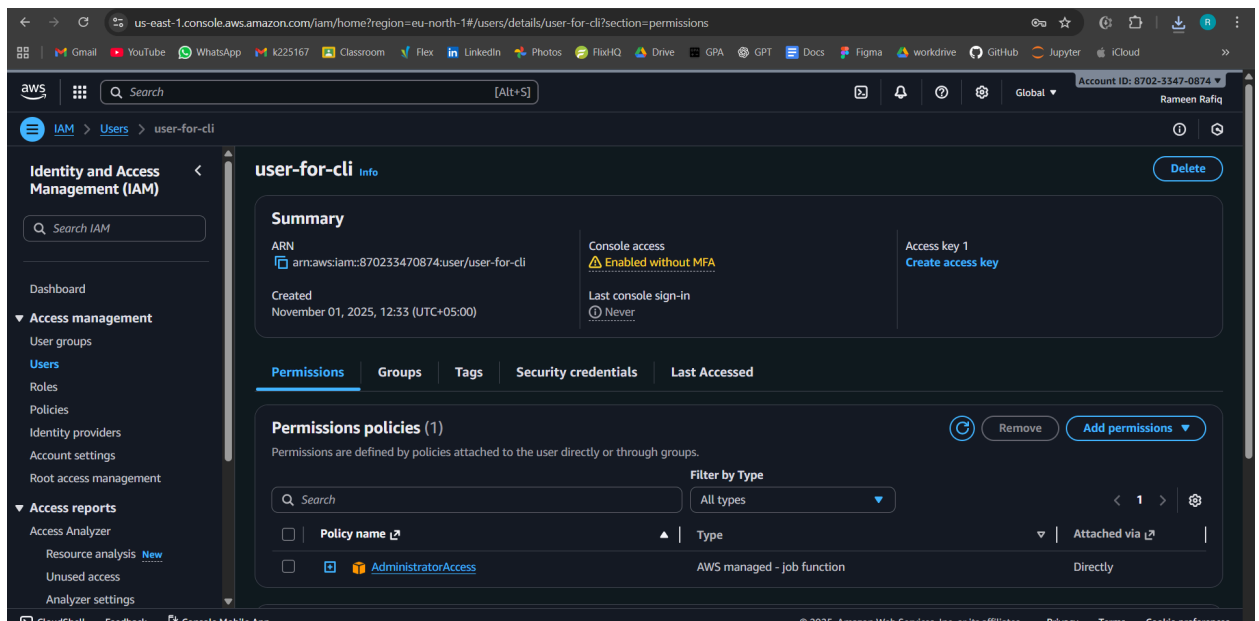
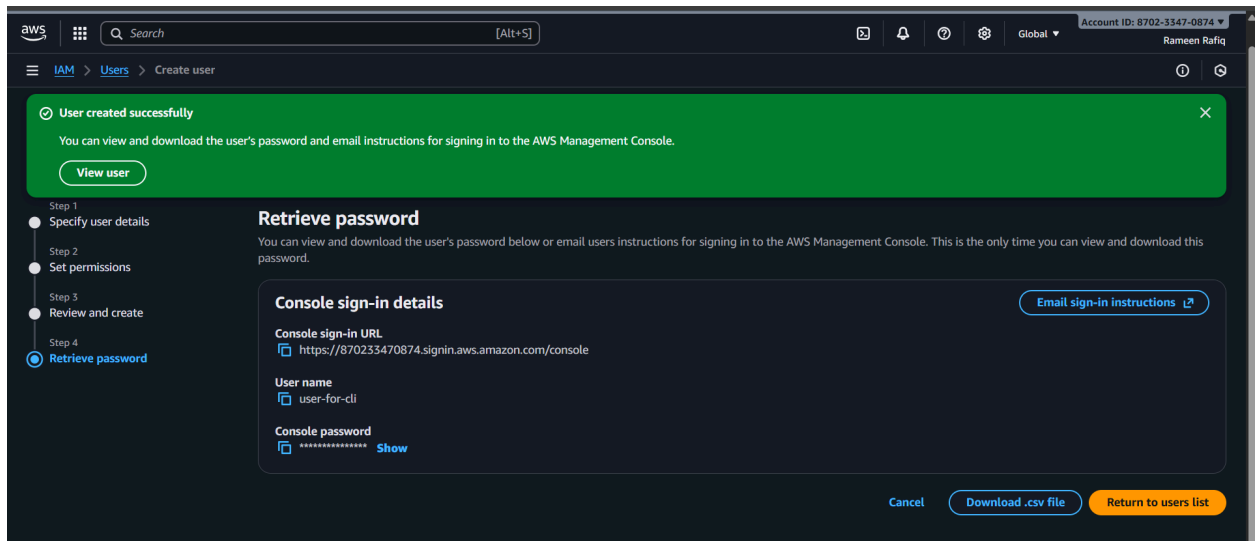
admin

Filter by Type

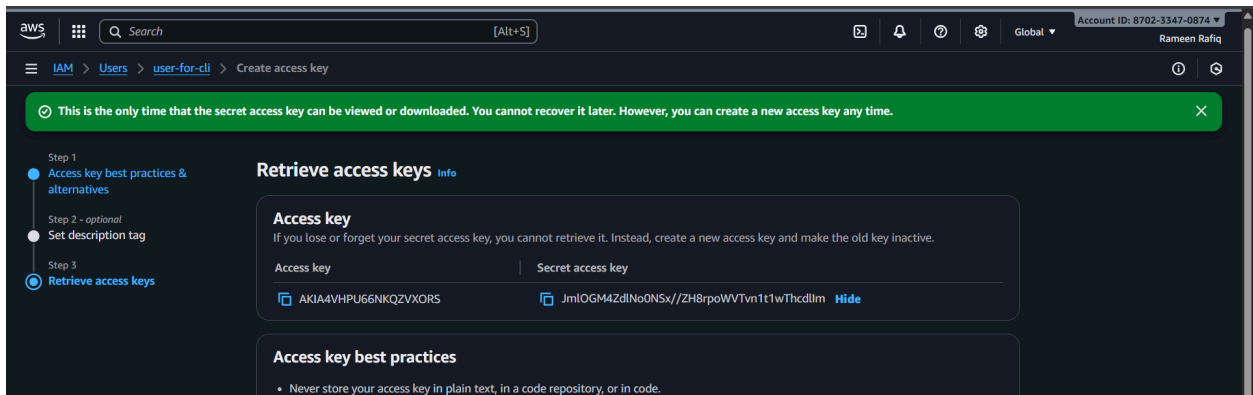
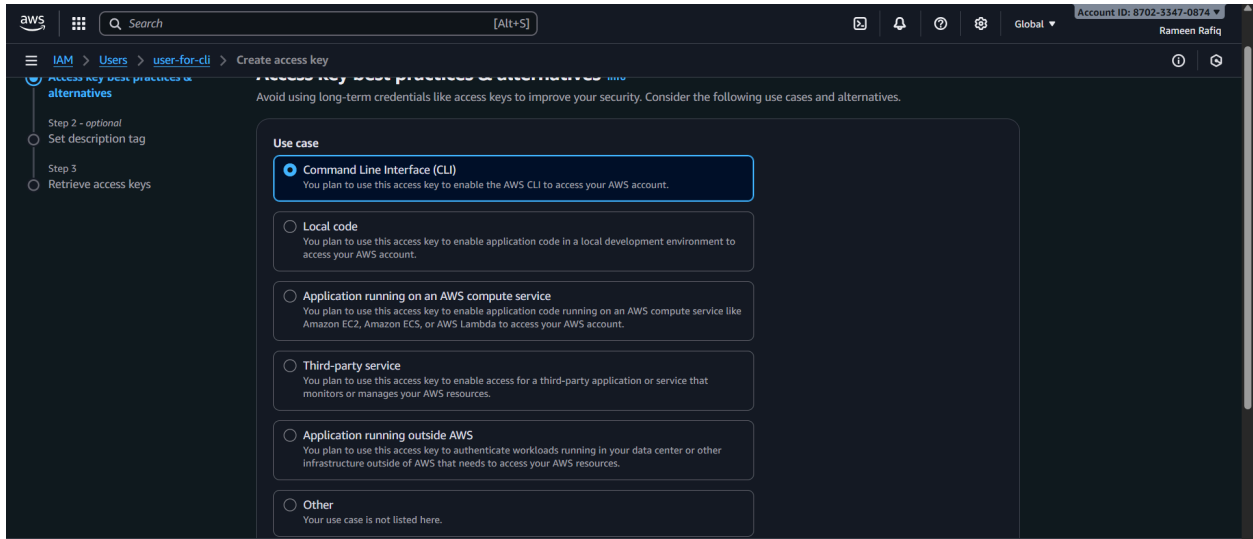
All types

53 matches

|                                     | Policy name                                             | Type                       | Attached entities |
|-------------------------------------|---------------------------------------------------------|----------------------------|-------------------|
| <input checked="" type="checkbox"/> | <a href="#">AdministratorAccess</a>                     | AWS managed - job function | 0                 |
| <input type="checkbox"/>            | <a href="#">AdministratorAccess-Amplify</a>             | AWS managed                | 0                 |
| <input type="checkbox"/>            | <a href="#">AdministratorAccess-AWSElasticBeanstalk</a> | AWS managed                | 0                 |



Note the create access key option

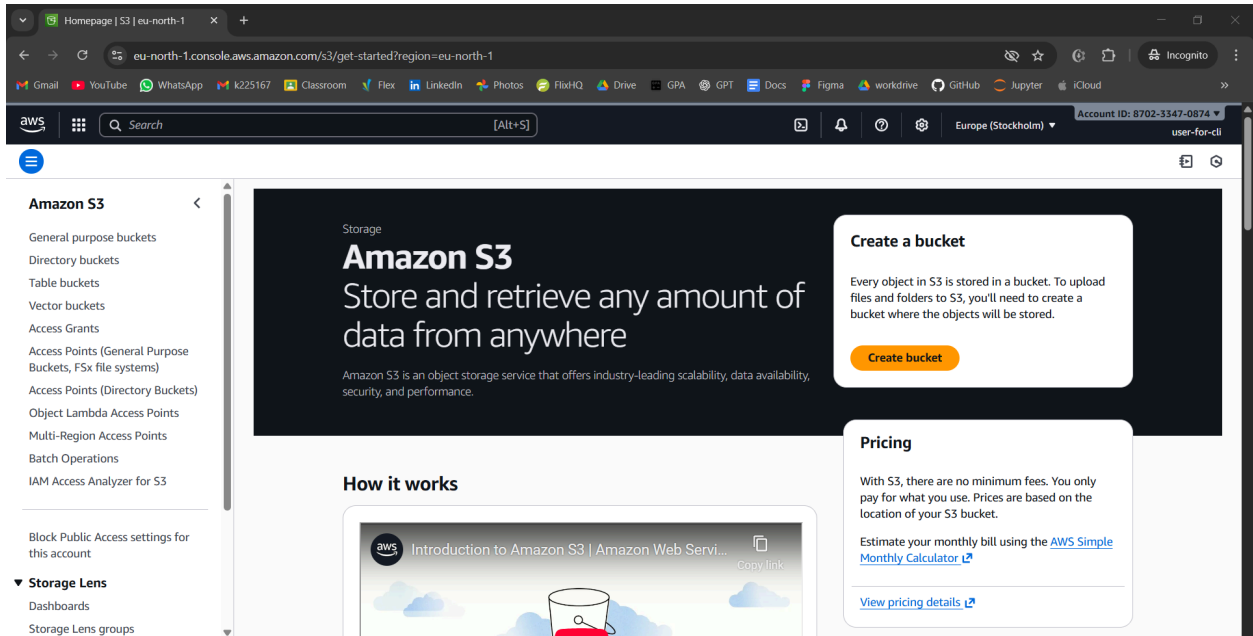


Access key: AKIA4VHPU66NKQZVXORS

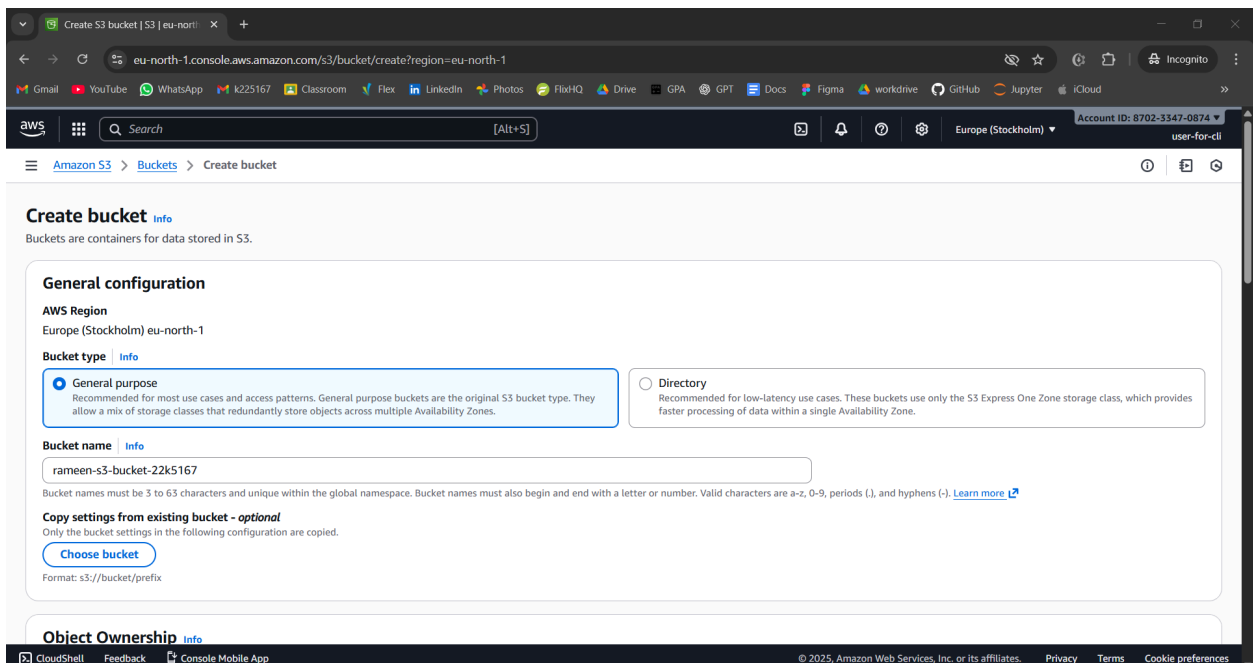
Secret access key: JmIOGM4ZdlNo0NSx//ZH8rpoWVTvn1t1wThcdllm

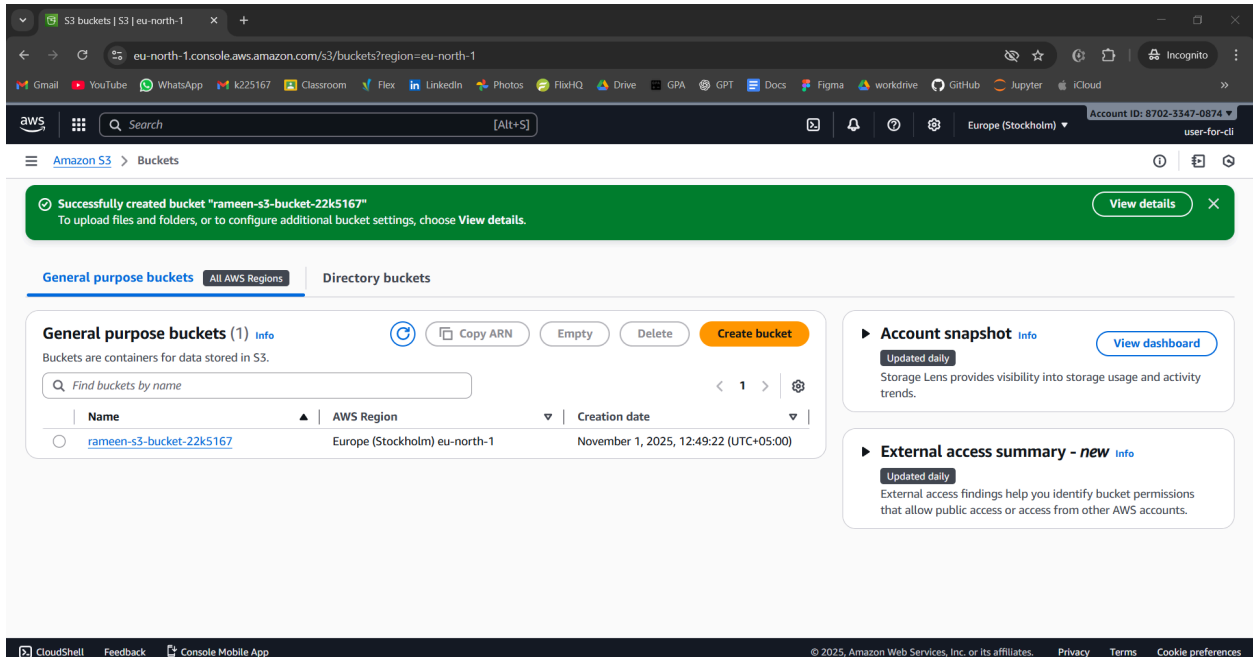
Now go to incognito and access the user just created

Go to s3

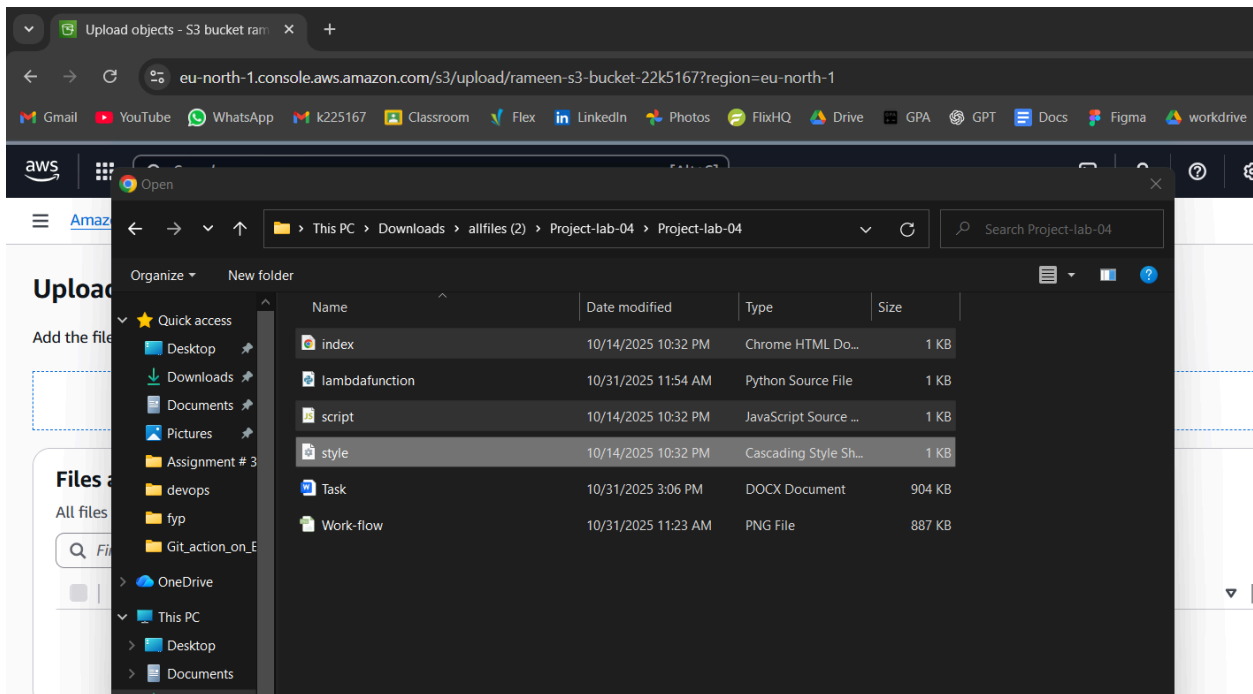


Now create a bucket

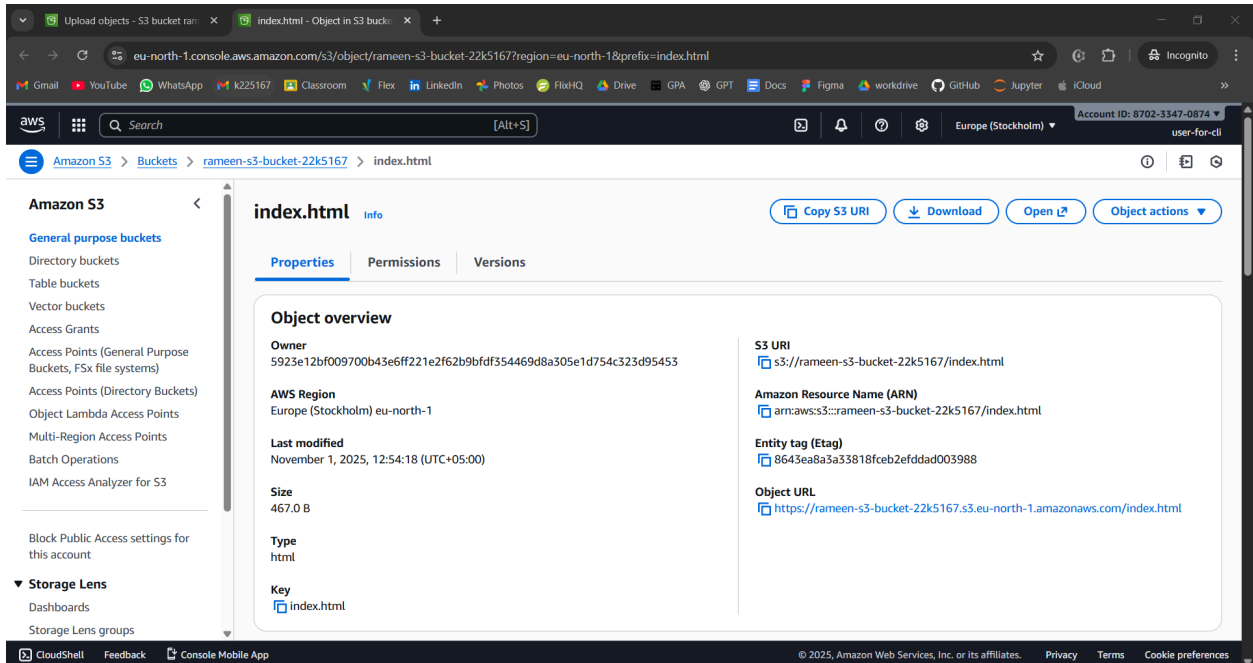




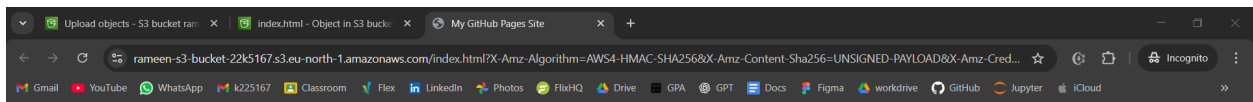
## Upload files in bucket



## Options for a file



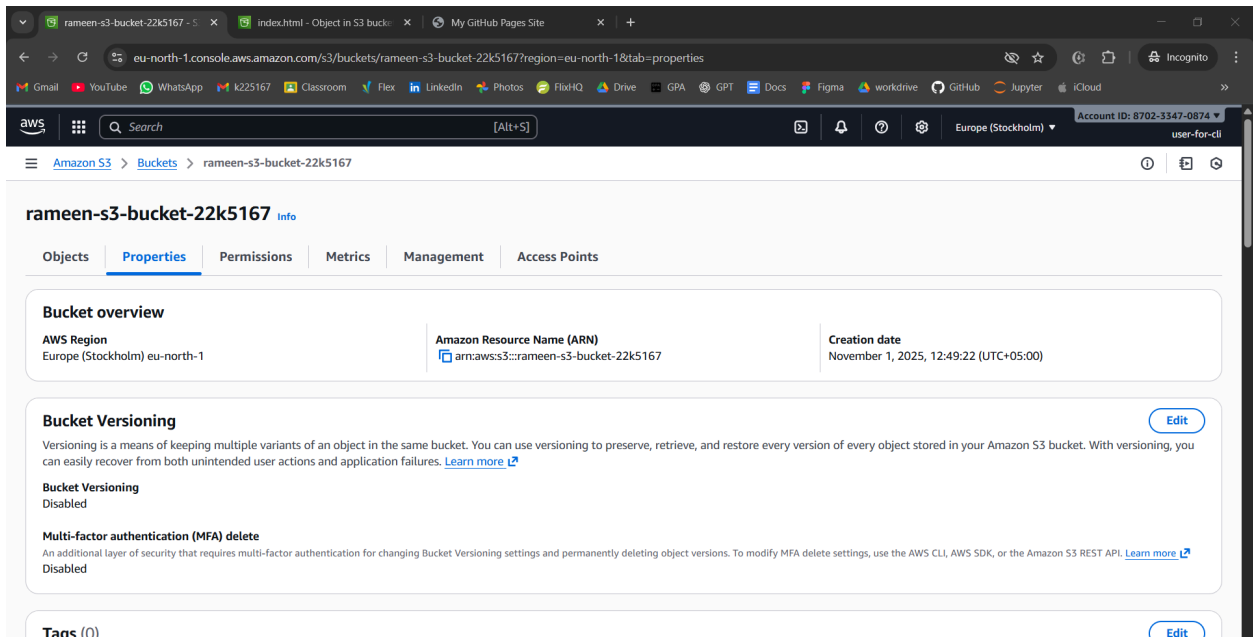
After opening the index file



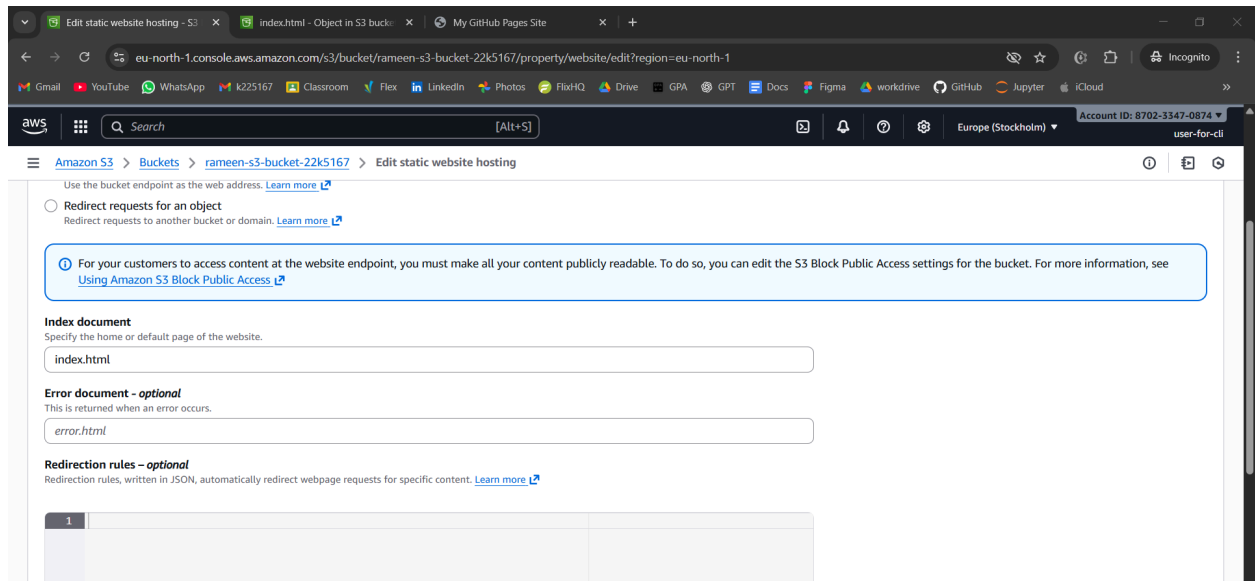
Welcome to My Website!

This site is hosted automatically using GitHub Actions

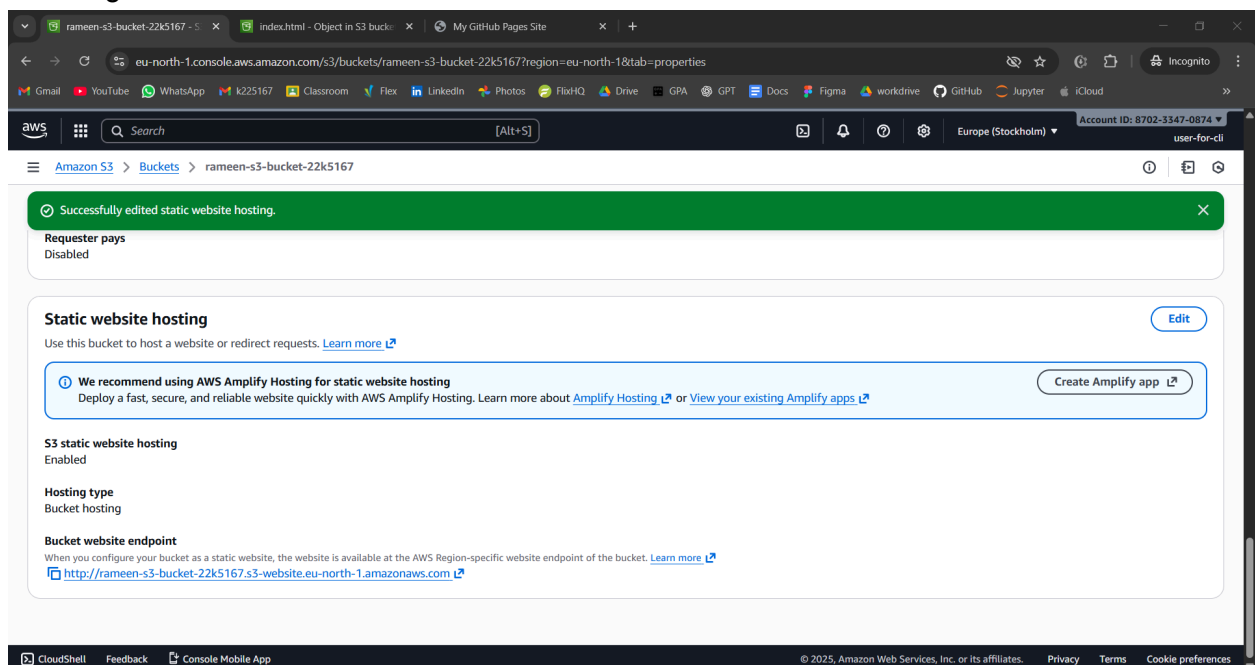
Click Me!



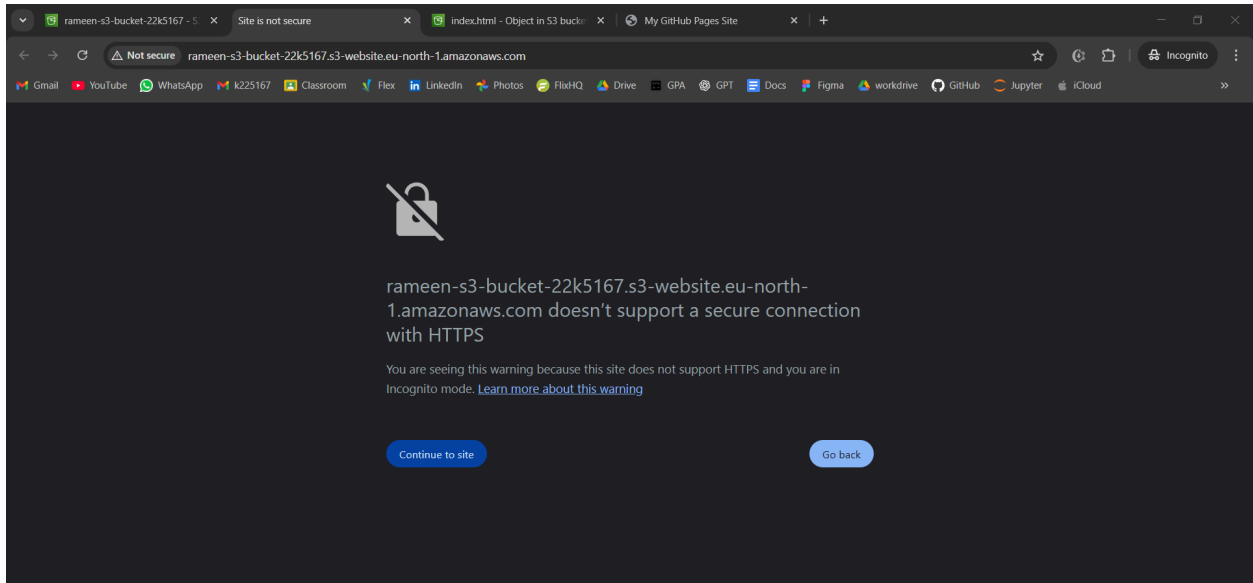
Come to end of page  
Click on edit



Url being shown at the end now

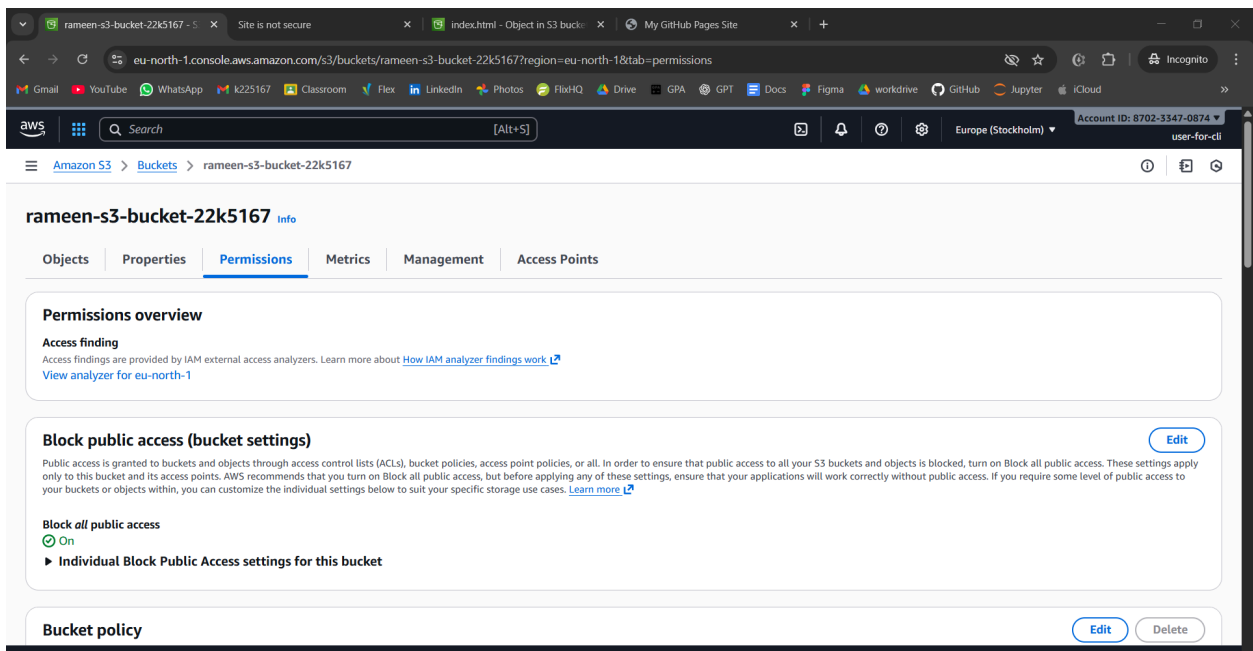


This shows error



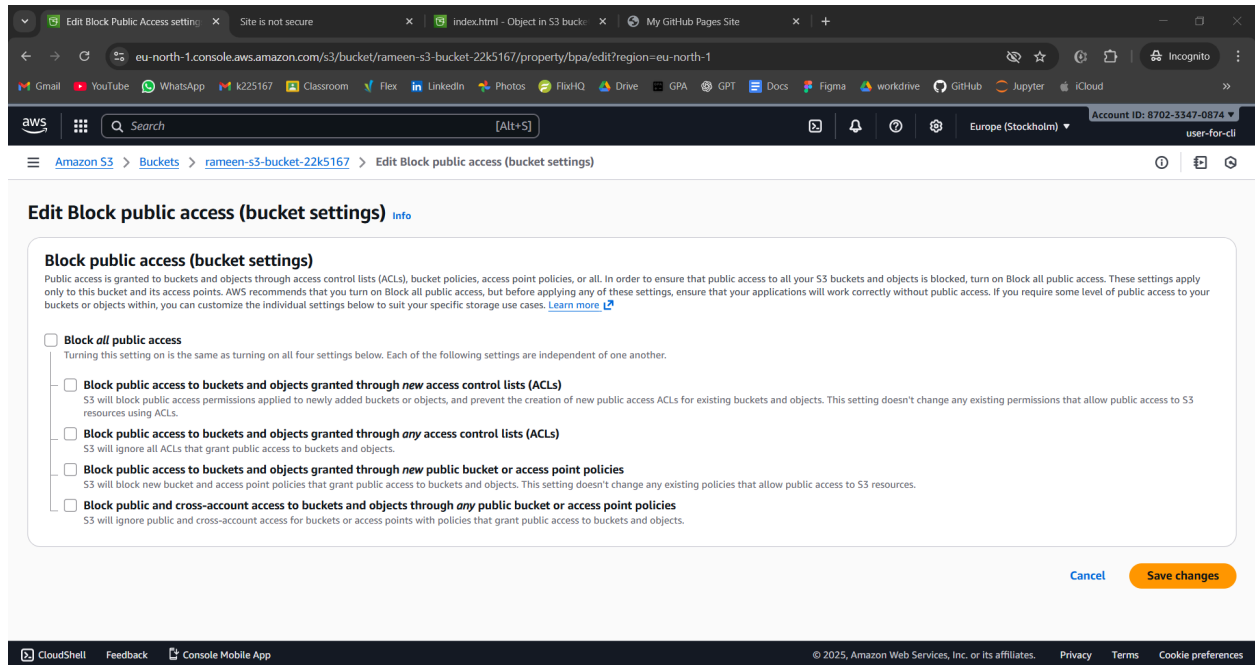
First we were just seeing the bucket. But then we are accessing the public url. So we need to give access

Now come here



Turn off block

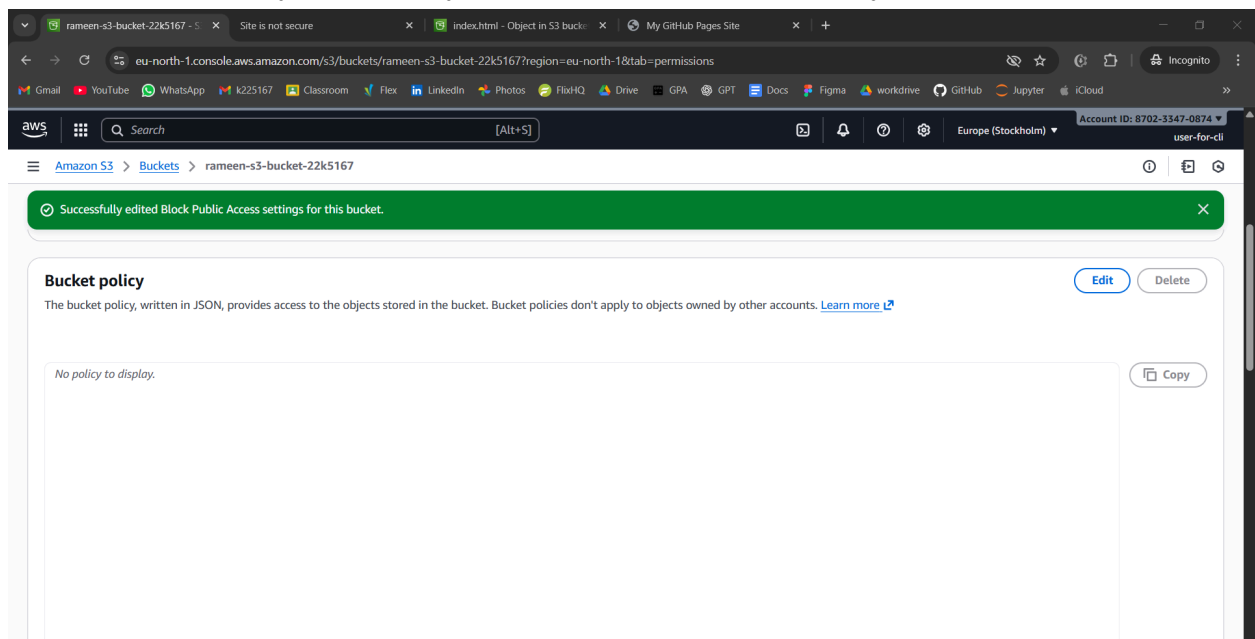


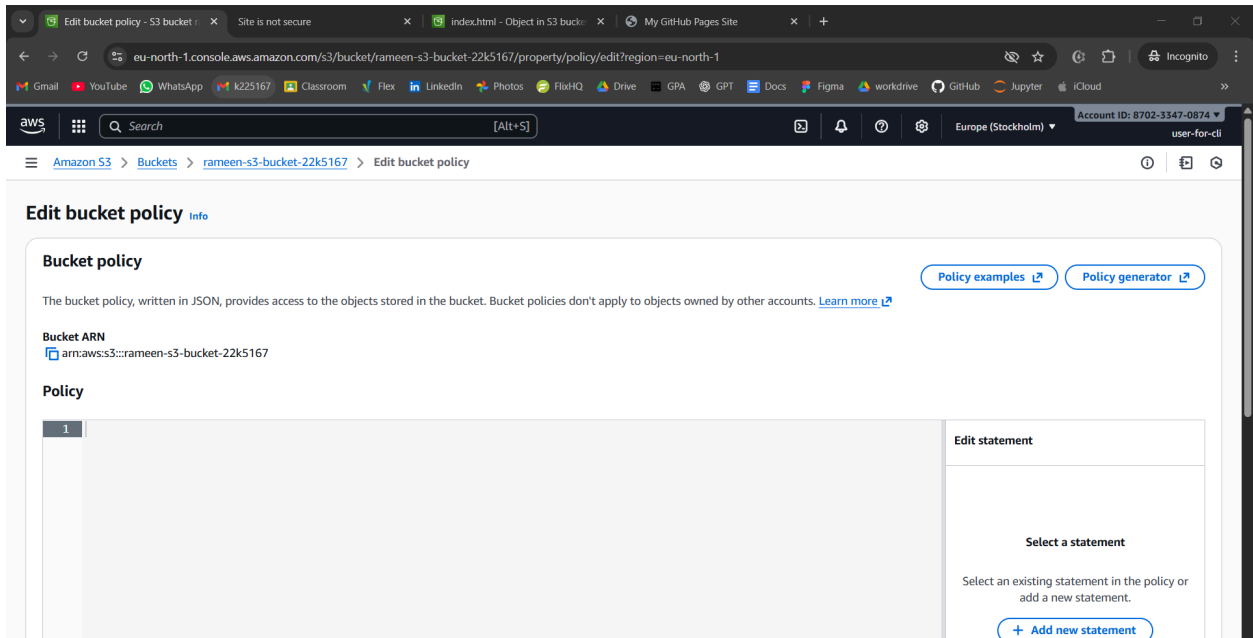


It still wont work, bec we just enabled to get permission, but not granted the permission. So grant permission now

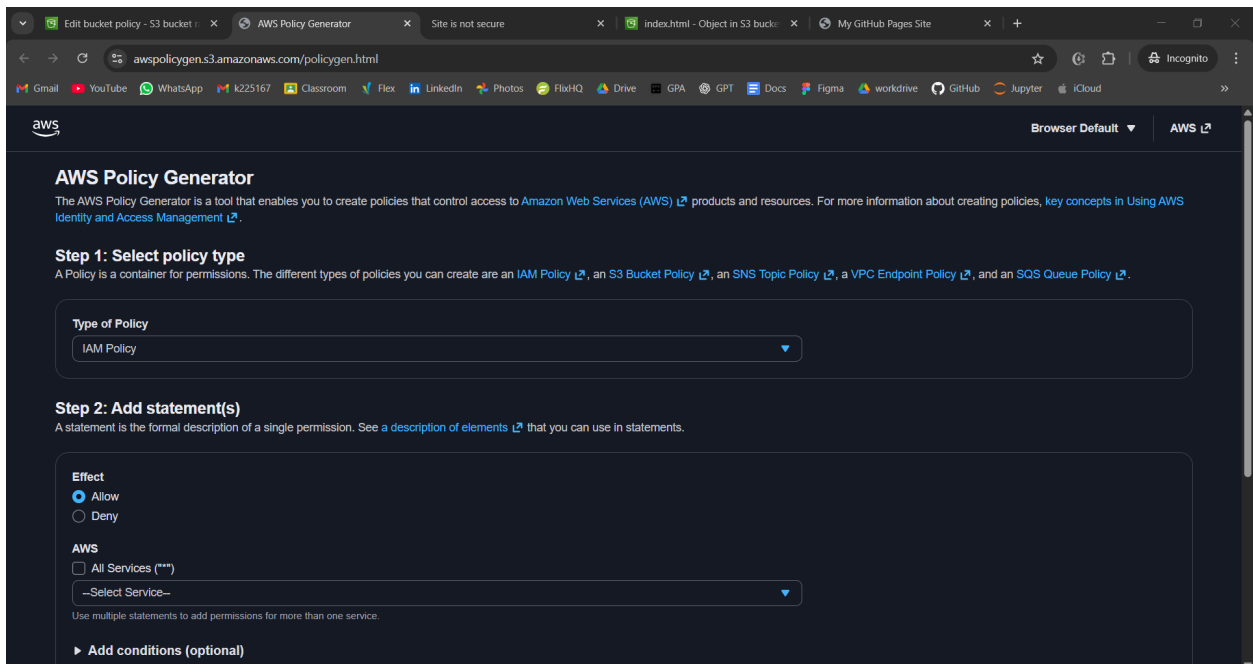
We need to grant permission here:

it acts like an extra layer of security. For this bucket we create policy ourself.





This interface will come:



Step 1: Select policy type

A Policy is a container for permissions. The different types of policies you can create are an [IAM Policy](#), an [S3 Bucket Policy](#), an [SNS Topic Policy](#), a [VPC Endpoint Policy](#), and an [SQS Queue Policy](#).

Type of Policy

S3 Bucket Policy

Step 2: Add statement(s)

A statement is the formal description of a single permission. See a [description of elements](#) that you can use in statements.

Effect

☒ Allow

☐ Deny

Principal

\*

Use a comma to separate multiple values.

Actions

☐ All Actions (\*\*\*)

--Select Actions--

GetObject

Copy the arn for bucket

Use a comma to separate multiple values.

Actions

☐ All Actions (\*\*\*)

--Select Actions--

GetObject

Amazon Resource Name (ARN)

☐ All Resources (\*\*\*)

arn:aws:s3:::rameen-s3-bucket-22k5167

ARN should follow the following format: arn:aws:s3:::{BucketName}/{Key Name}. Use a comma to separate multiple values.

► Add conditions (optional)

Add Statement

Add statement

ARN should follow the following format: arn:aws:s3:::{BucketName}/{Key Name}. Use a comma to separate multiple values.

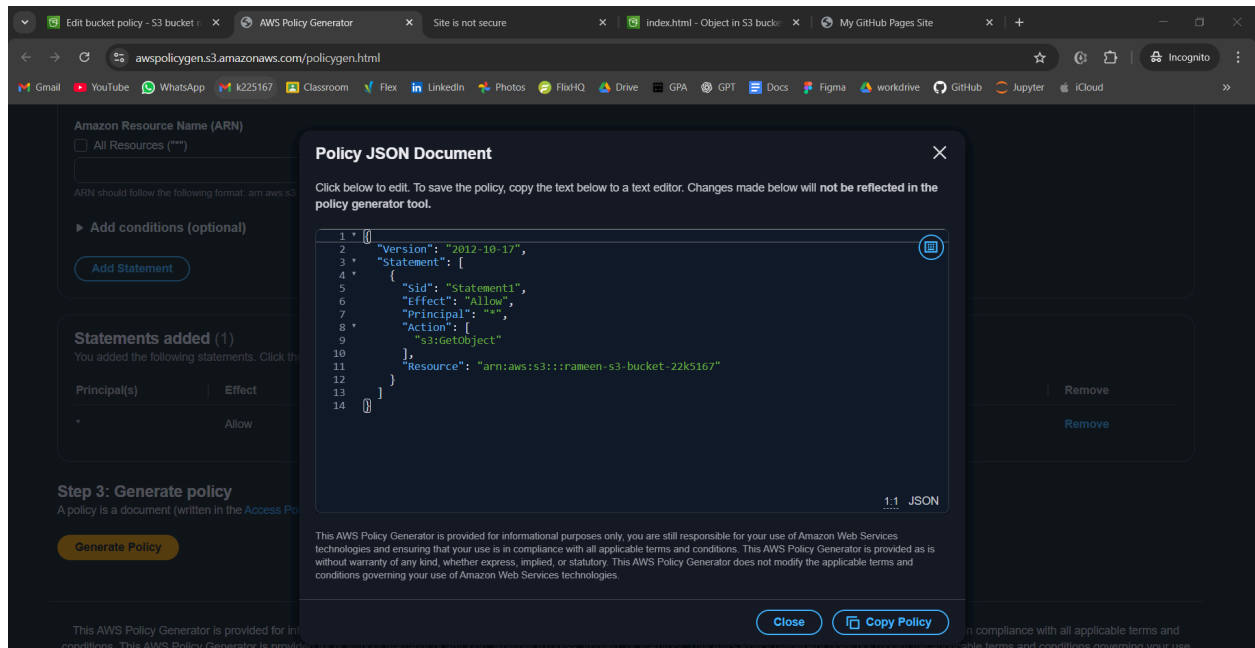
► Add conditions (optional)

Add Statement

Statements added (1)

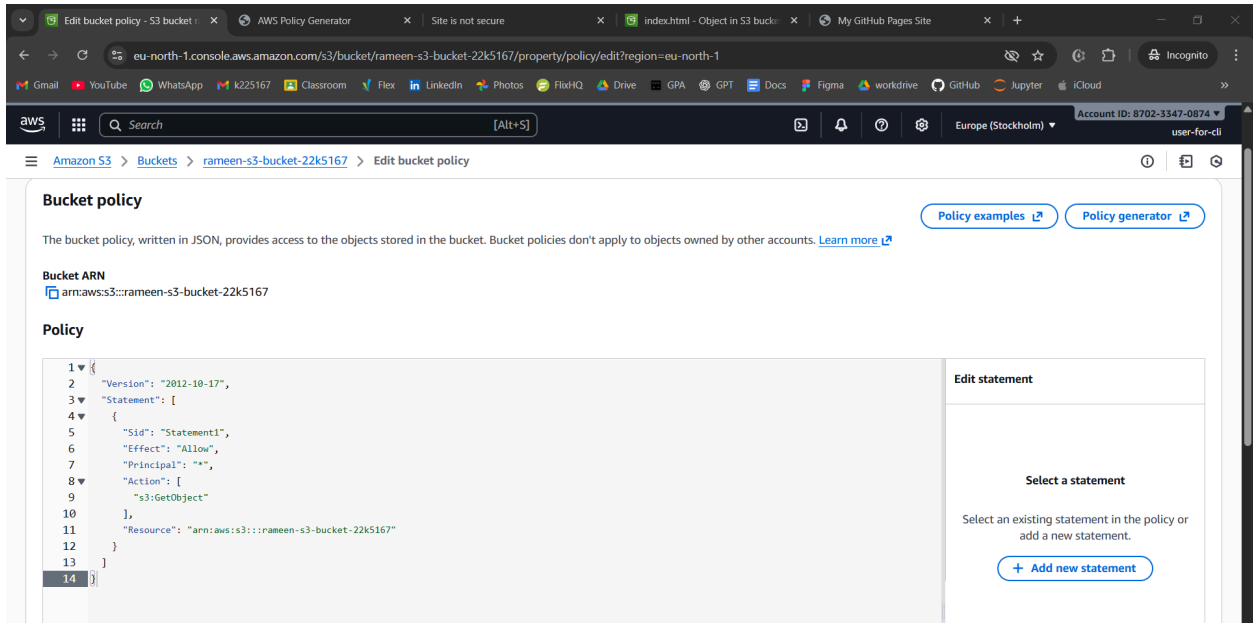
You added the following statements. Click the button below to Generate a policy.

| Principal(s) | Effect | Action       | Resource(s)                           | Condition(s) | Remove |
|--------------|--------|--------------|---------------------------------------|--------------|--------|
| *            | Allow  | s3:GetObject | arn:aws:s3:::rameen-s3-bucket-22k5167 | None         | Remove |

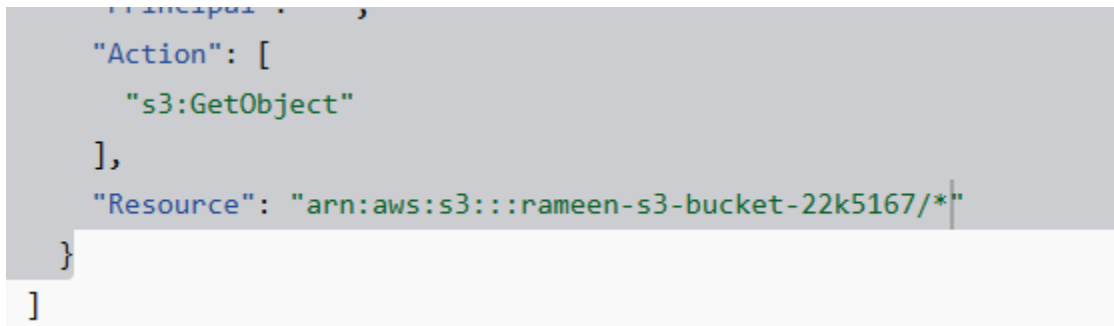


```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "Statement1",
      "Effect": "Allow",
      "Principal": "*",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": "arn:aws:s3:::rameen-s3-bucket-22k5167"
    }
  ]
}
```

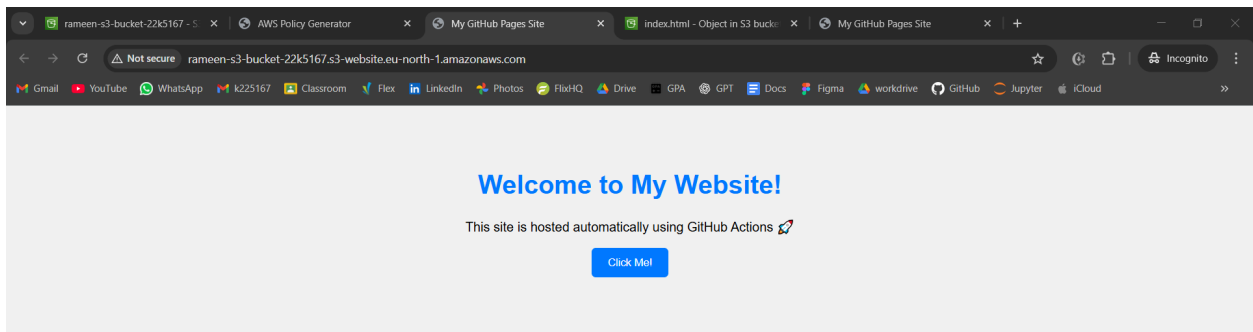
Paste the policy here:



Change this



Now go to site and continue to page



Lambda, sns

Now configuring sns

Simple Notification Service | eu | AWS Policy Generator | My GitHub Pages Site | index.html - Object in S3 bucket | My GitHub Pages Site | +

eu-north-1.console.aws.amazon.com/sns/v3/home?region=eu-north-1#/homepage

Gmail YouTube WhatsApp k225167 Classroom Flex LinkedIn Photos FlickrHQ Drive GPA GPT Docs Figma workdrive GitHub Jupyter iCloud

aws Search [Alt+S] Europe (Stockholm) Account ID: 8702-3347-0874 user-for-cli

**New Feature**  
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Application Integration

# Amazon Simple Notification Service

Pub/sub messaging for microservices and serverless applications.

Amazon SNS is a highly available, durable, secure, fully managed pub/sub messaging service that enables you to decouple microservices, distributed systems, and event-driven serverless applications. Amazon SNS provides topics for high-throughput, push-based, many-to-many messaging.

### Create topic

**Topic name**  
A topic is a message channel. When you publish a message to a topic, it fans out the message to all subscribed endpoints.

**Next step**

[Start with an overview](#)

### Pricing

Amazon SNS has no upfront costs. You pay based

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Create topic | Topics | Simple N | AWS Policy Generator | My GitHub Pages Site | index.html - Object in S3 bucket | My GitHub Pages Site | +

eu-north-1.console.aws.amazon.com/sns/v3/home?region=eu-north-1#/create-topic

Gmail YouTube WhatsApp k225167 Classroom Flex LinkedIn Photos FlickrHQ Drive GPA GPT Docs Figma workdrive GitHub Jupyter iCloud

aws Search [Alt+S] Europe (Stockholm) Account ID: 8702-3347-0874 user-for-cli

Amazon SNS > Topics > Create topic

## Details

**Type** [Info](#)  
Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

**Name**

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (\_).

**Display name - optional** [Info](#)  
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

Maximum 100 characters.

aws Search [Alt+S] Europe (Stockholm) Account ID: 8702-3347-0874 user-for-eli

Amazon SNS > Topics > Create topic

**Display name - optional** [info](#)  
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

my-notification-for-s3  
Maximum 100 characters.

my-notification | Topics | Simpli | AWS Policy Generator | My GitHub Pages Site | index.html - Object in S3 bucket | My GitHub Pages Site

eu-north-1.console.aws.amazon.com/sns/v3/home?region=eu-north-1#/topic/arn:aws:sns:eu-north-1:870233470874:my-notification

Gmail YouTube WhatsApp k225167 Classroom Flex LinkedIn Photos FluxHQ Drive GPA GPT Docs Figma workdrive GitHub Jupyter iCloud

aws Search [Alt+S] Europe (Stockholm) Account ID: 8702-3347-0874 user-for-eli

Amazon SNS > Topics > my-notification

**Amazon SNS**

- Dashboard
- Topics
- Subscriptions
- ▼ Mobile
  - Push notifications
  - Text messaging (SMS)

**Topic my-notification created successfully.**  
You can create subscriptions and send messages to them from this topic. [Publish message](#) ✕

**my-notification** [Edit](#) [Delete](#) [Publish message](#)

**Details**

|                                                                   |                                               |
|-------------------------------------------------------------------|-----------------------------------------------|
| <b>Name</b><br>my-notification                                    | <b>Display name</b><br>my-notification-for-s3 |
| <b>ARN</b><br>arn:aws:sns:eu-north-1:870233470874:my-notification | <b>Topic owner</b><br>870233470874            |
| <b>Type</b><br>Standard                                           |                                               |

[Subscriptions](#) [Access policy](#) [Data protection policy](#) [Delivery policy \(HTTP/S\)](#) [Delivery status logging](#) [Encryption](#) [Tags](#)

**Subscriptions (0)** [Edit](#) [Delete](#) [Request confirmation](#) [Confirm subscription](#) [Create subscription](#)

< 1 > ⚙

Click on create subscription

Create subscription | Subscrip... AWS Policy Generator My GitHub Pages Site index.html - Object in S3 bucke... My GitHub Pages Site +

eu-north-1.console.aws.amazon.com/sns/v3/home?region=eu-north-1#/create-subscription

Amazon SNS > Subscriptions > Create subscription

### Create subscription

**Details**

**Topic ARN**

arn:aws:sns:eu-north-1:870233470874:my-notification

**Protocol**

The type of endpoint to subscribe

Email

**Endpoint**

An email address that can receive notifications from Amazon SNS.

rameenrafiq27@gmail.com

After your subscription is created, you must confirm it. [Info](#)

**Subscription filter policy - optional** [Info](#)

This policy filters the messages that a subscriber receives.

Subscription: 5673e740-d0f1-473b-82c4-ebddf2c3da37

eu-north-1.console.aws.amazon.com/sns/v3/home?region=eu-north-1#/subscription/arn:aws:sns:eu-north-1:870233470874:my-notification:5673e740-d0f1-473b-82c4-ebddf2c3da37

Amazon SNS > Topics > my-notification > Subscription: 5673e740-d0f1-473b-82c4-ebddf2c3da37

**Amazon SNS**

Dashboard

Topics

**Subscriptions**

▼ Mobile

Push notifications

Text messaging (SMS)

Subscription to my-notification created successfully.

The ARN of the subscription is arn:aws:sns:eu-north-1:870233470874:my-notification:5673e740-d0f1-473b-82c4-ebddf2c3da37.

**Subscription: 5673e740-d0f1-473b-82c4-ebddf2c3da37**

EditDelete

**Details**

**ARN**

arn:aws:sns:eu-north-1:870233470874:my-notification:5673e740-d0f1-473b-82c4-ebddf2c3da37

**Endpoint**

rameenrafiq27@gmail.com

**Topic**

[my-notification](#)

**Subscription Principal**

arn:aws:iam::870233470874:user/user-for-cli

**Status**

Pending confirmation

**Protocol**

EMAIL

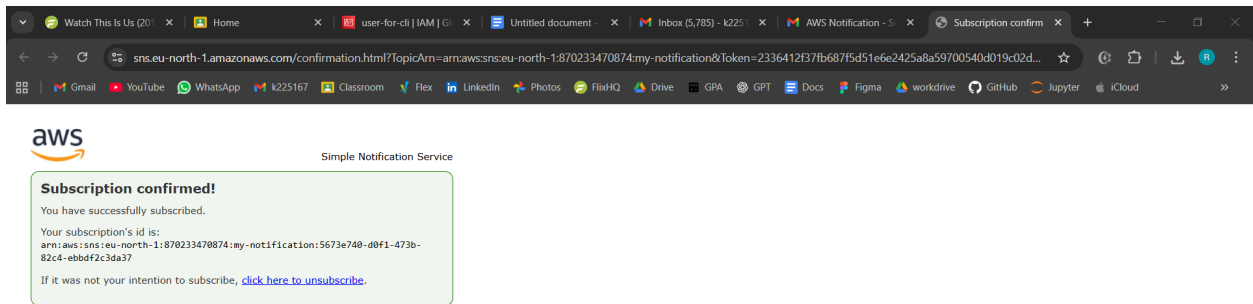
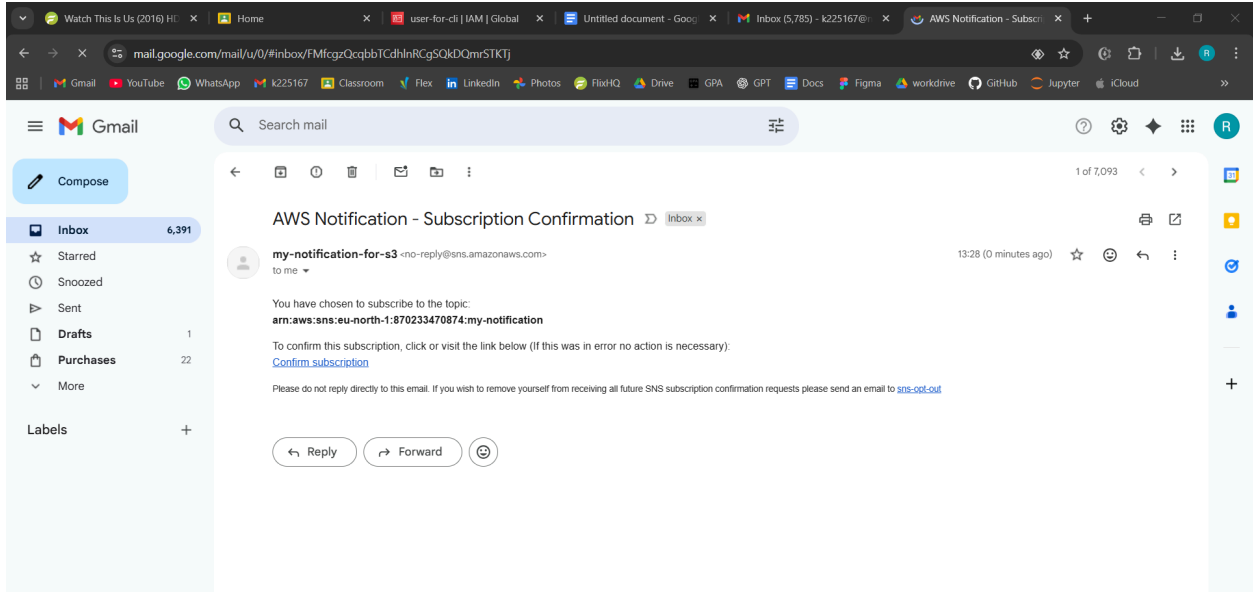
**Subscription filter policy**

Redrive policy (dead-letter queue)

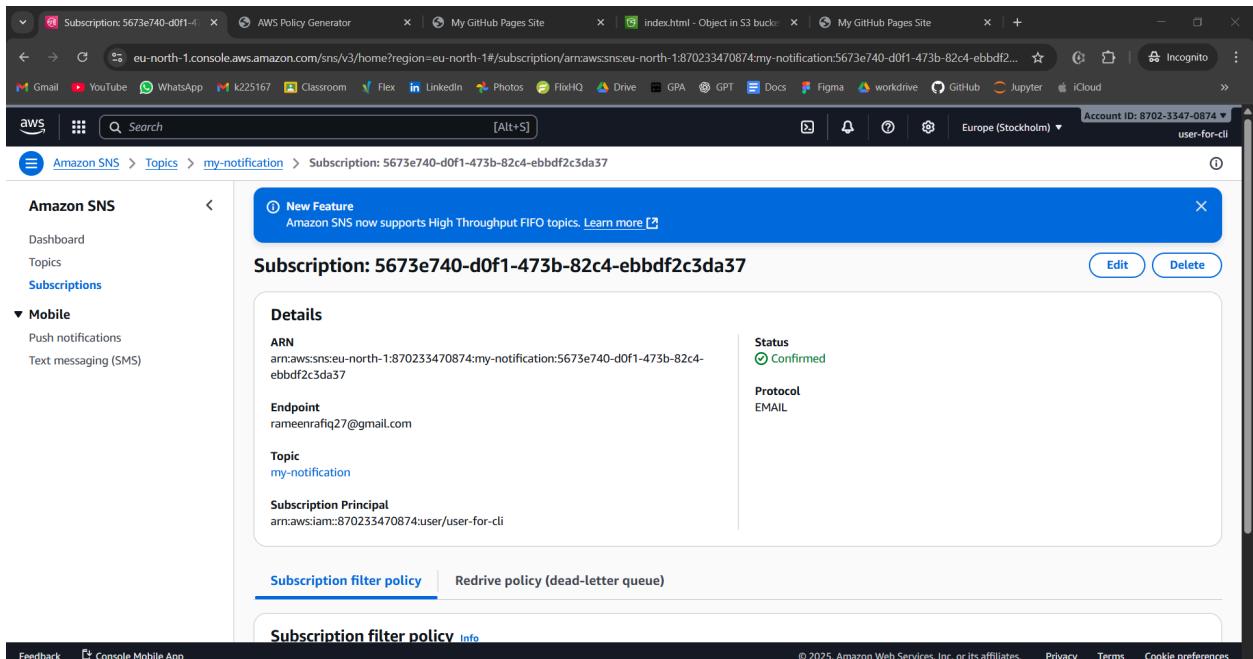
**Subscription filter policy** [Info](#)

Status is pending





Status is now confirmed



Now we create lambda function and connect them

**Create function** [Info](#)

Choose one of the following options to create your function.

☒ **Author from scratch**  
Start with a simple Hello World example.

☐ **Use a blueprint**  
Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**  
Select a container image to deploy for your function.

**Basic information**

**Function name**  
Enter a name that describes the purpose of your function.  
  
Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (\_).

**Runtime** [Info](#)  
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

**Architecture** [Info](#)  
Choose the instruction set architecture you want for your function code.  
☐ arm64  
☒ x86\_64

Successfully created the function **snsfunction**. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

**snsfunction** [Throttle](#) [Copy ARN](#) [Actions](#)

[Export to Infrastructure Composer](#) [Download](#)

**Function overview** [Info](#)

[Diagram](#) [Template](#)

**snsfunction**

Layers (0)

[+ Add trigger](#) [+ Add destination](#)

**Description**  
-

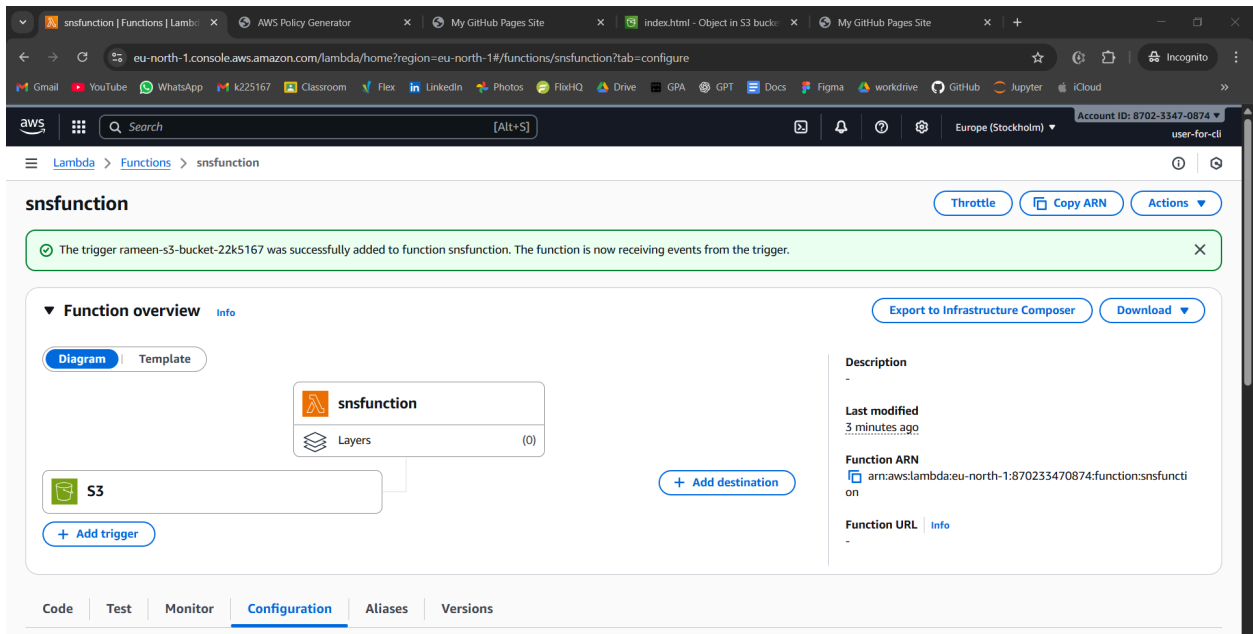
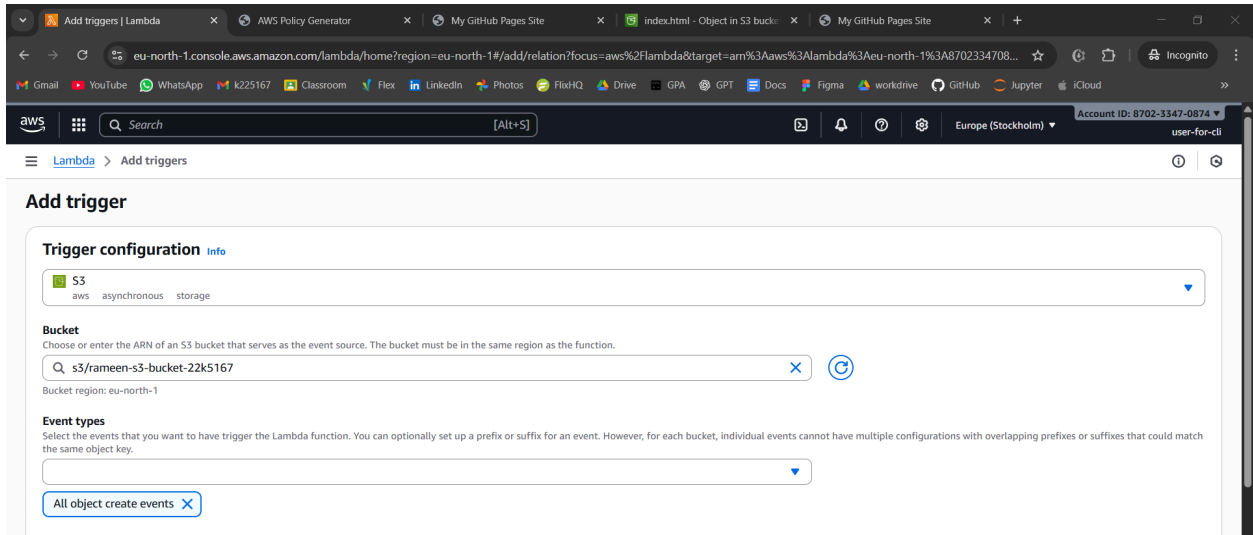
**Last modified**  
14 seconds ago

**Function ARN**  
[arn:aws:lambda:eu-north-1:870233470874:function:snsfunction](#)

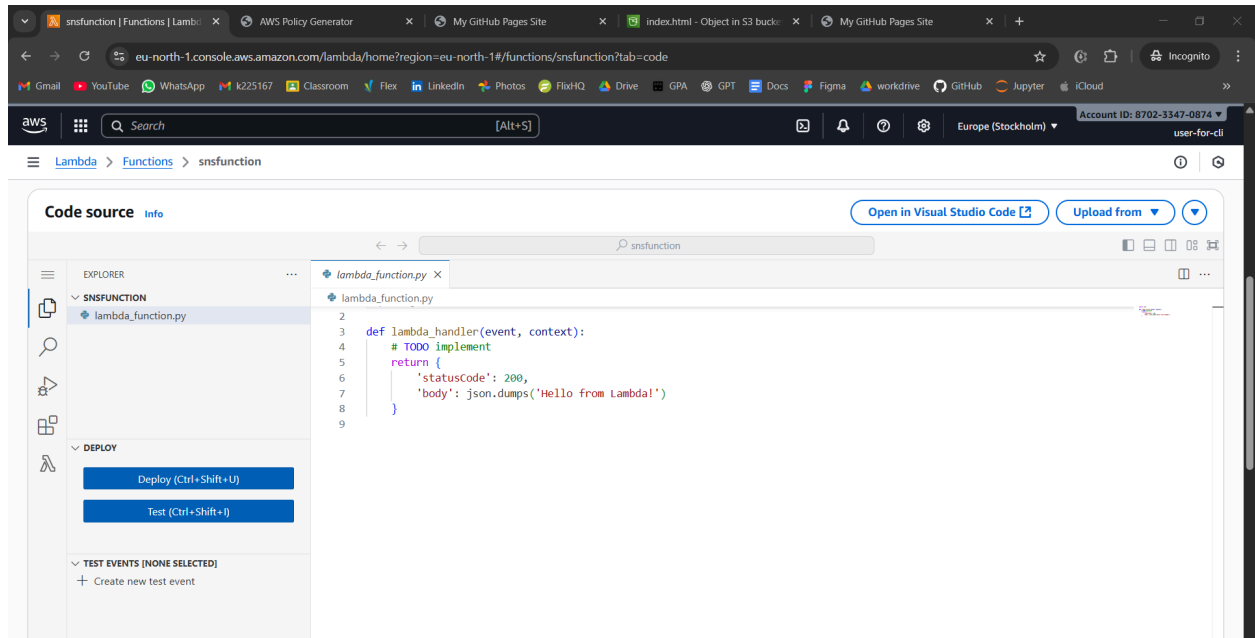
**Function URL** [Info](#)  
-

[Code](#) [Test](#) [Monitor](#) [Configuration](#) [Aliases](#) [Versions](#)

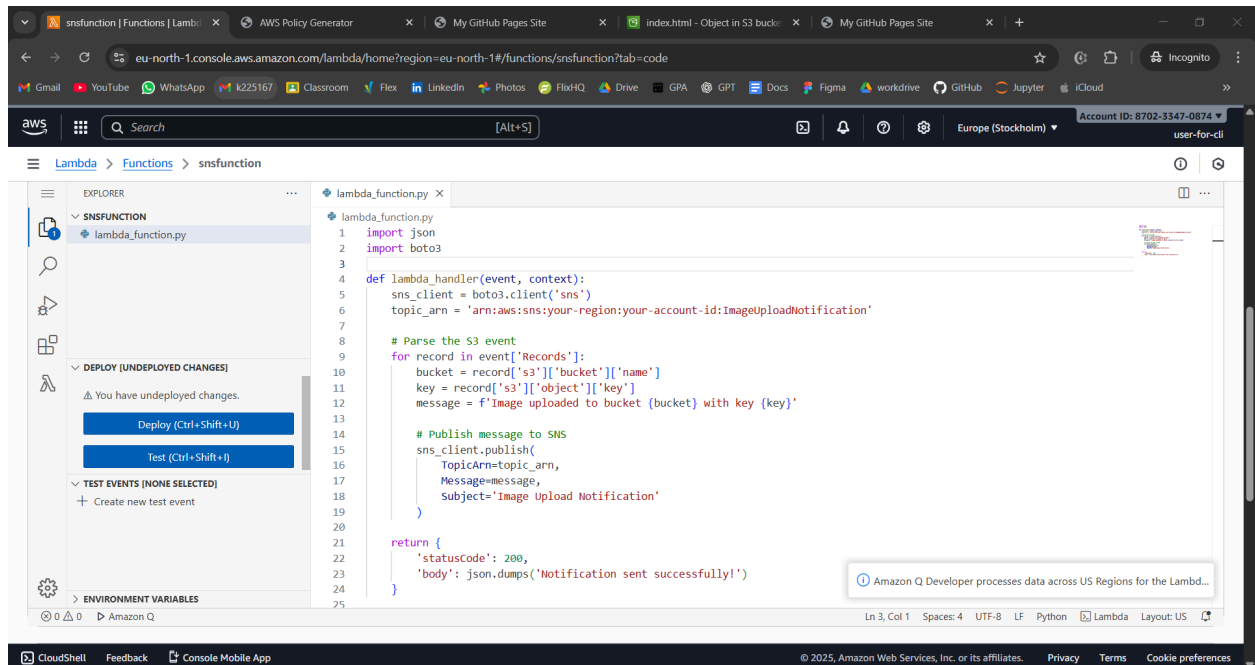
CloudShell Feedback Console Mobile App © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



Click on layers  
Go to code option



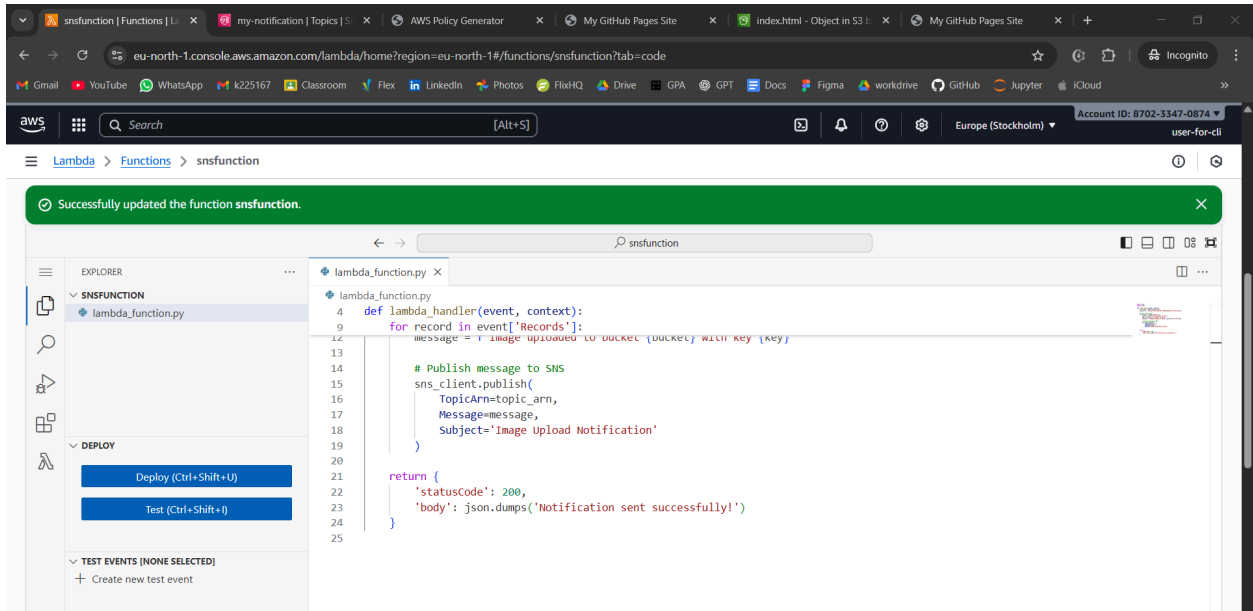
Open the lambda function file in folder  
And paste it



Change the topic

```
topic_arn = 'arn:aws:sns:eu-north-1:870233470874:my-notification'
```

We use json bec we are coding in python  
Saves as name and prints message  
Then deploy

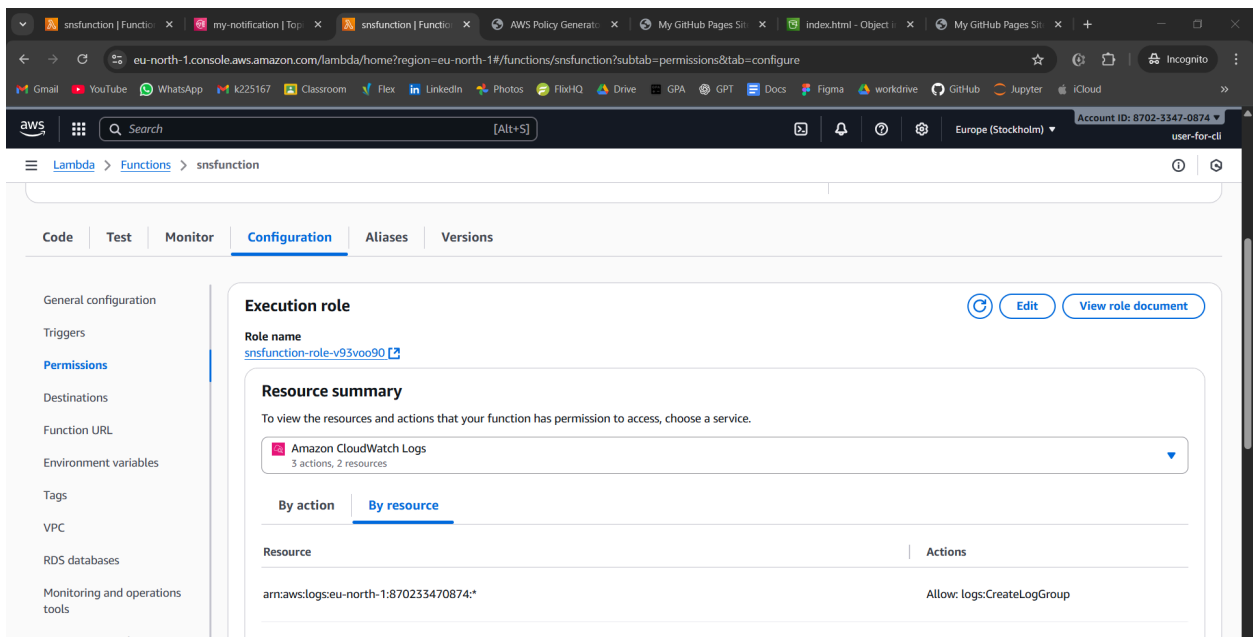


But notification still wont come bec permission not given.

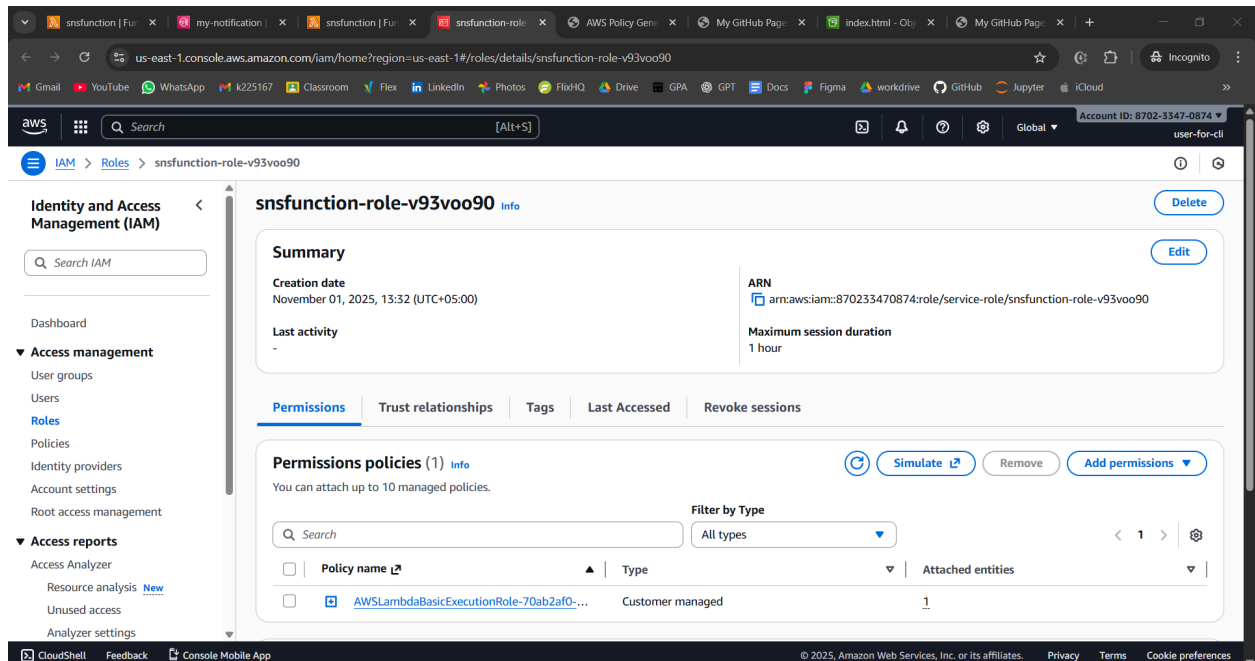
In aws if one service communicates with other, service also needs permission even though it is in same zone. Permissions also for services.

Roles can also be made for services, not only users

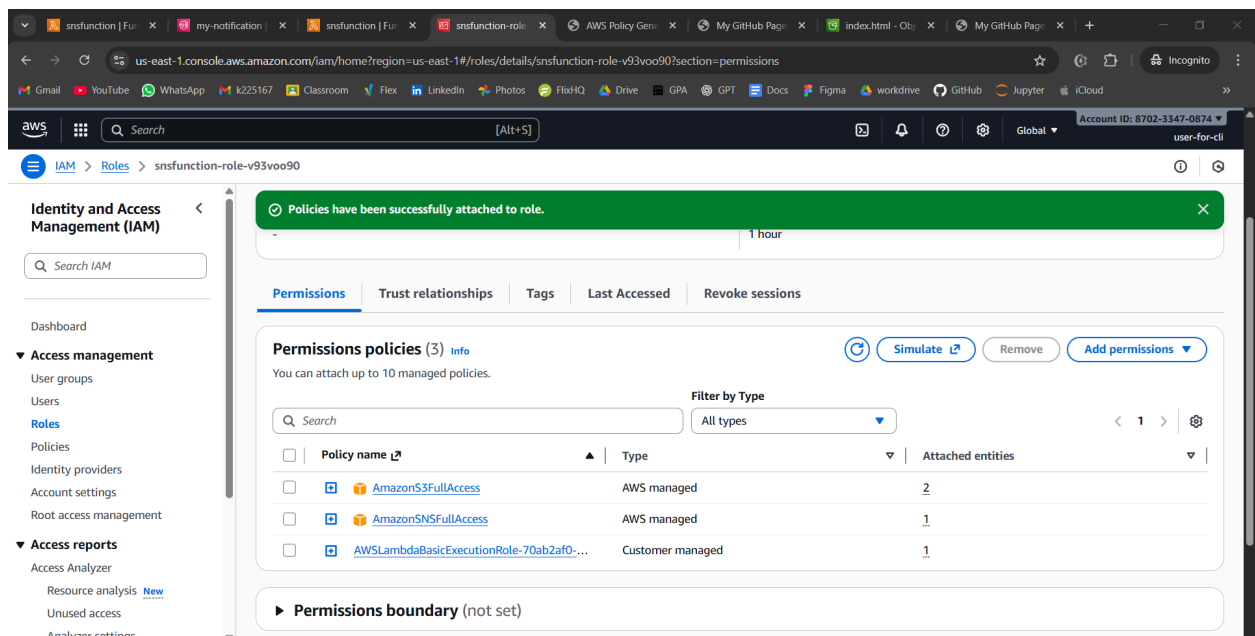
Now we have to give lambda permission to access s3 bucket and sns as well.



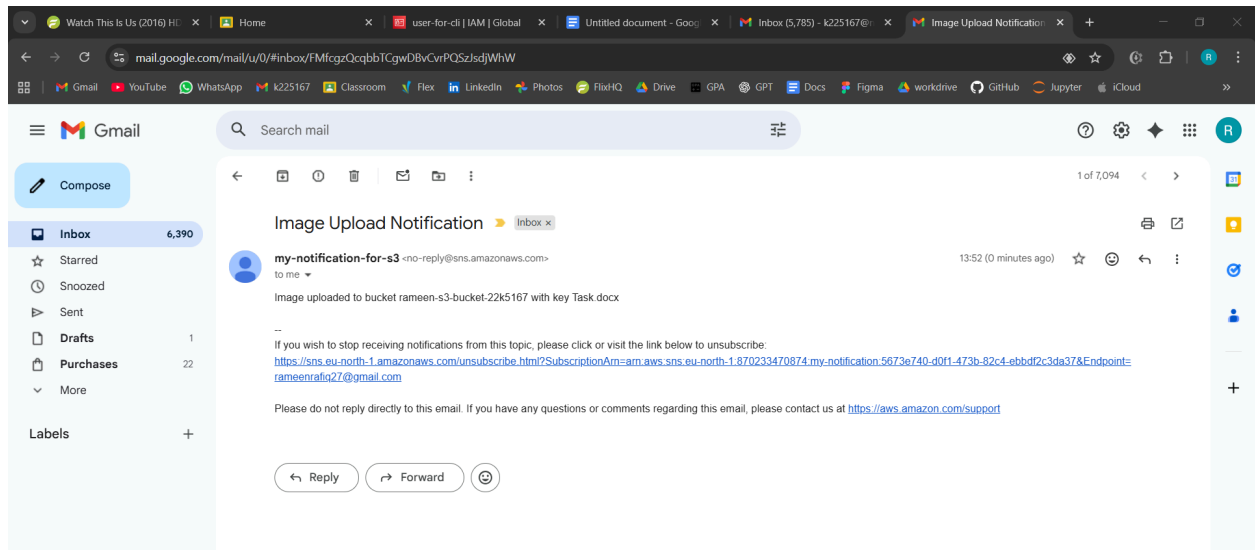
Click on role name



Now we add permission for s3 and sns

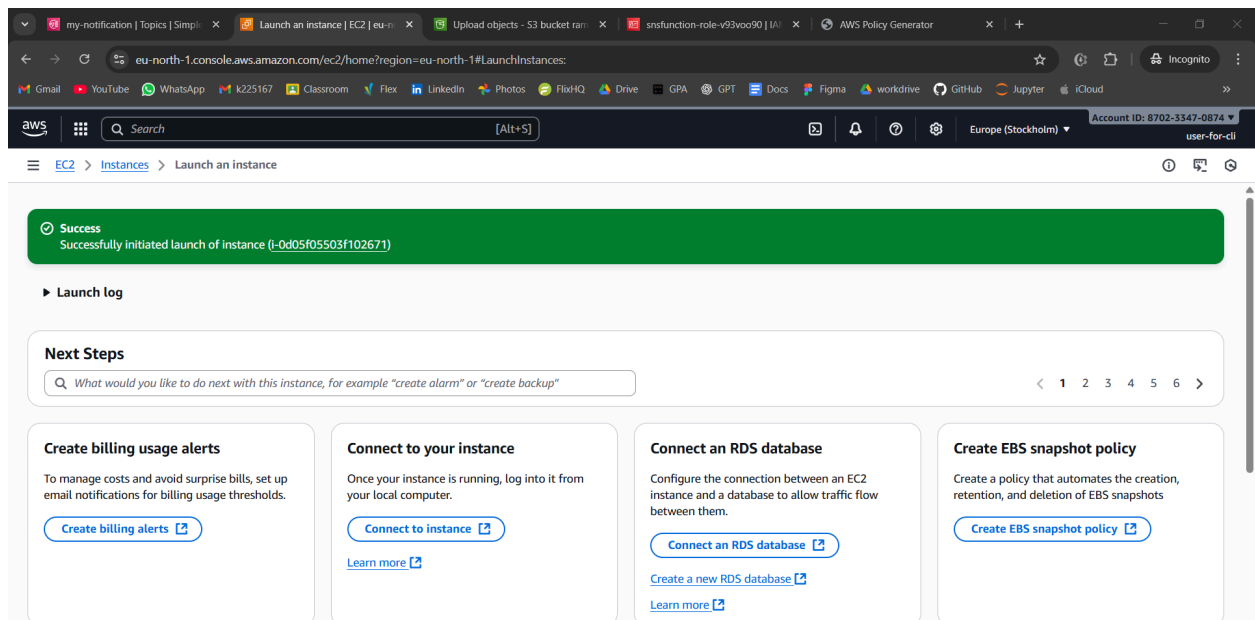


Upload any file to s3 and check email



## Second activity

### Create ec2



Then connect  
Now run all commands

```
inflatng: aws/dist/prompt toolkit-3.0.51.dist-info/WHEEL
inflatng: aws/dist/prompt toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
inflatng: aws/dist/prompt toolkit-3.0.51.dist-info/licenses/LICENSE
inflatng: aws/dist/wheel-0.45.1.dist-info/INSTALLER
inflatng: aws/dist/wheel-0.45.1.dist-info/WHEEL
inflatng: aws/dist/wheel-0.45.1.dist-info/METADATA
inflatng: aws/dist/wheel-0.45.1.dist-info/RECORD
inflatng: aws/dist/wheel-0.45.1.dist-info/direct_url.json
inflatng: aws/dist/wheel-0.45.1.dist-info/LICENSE.txt
inflatng: aws/dist/wheel-0.45.1.dist-info/REQUESTED
inflatng: aws/dist/wheel-0.45.1.dist-info/entry_points.txt
ubuntu@ip-172-31-25-174:~$ unzip awscli2.zip
Archive:  awscli2.zip
replace aws/README.md? [yes, [no], [All], [None], [r]ename: ^X
error:  invalid response []
replace aws/README.md? [yes, [no], [All], [None], [r]ename: ^Cubuntu@ip-172-31-25-174:~$ n
n: command not found
ubuntu@ip-172-31-25-174:~$ sudo ./aws/install
You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-25-174:~$ sudo ./aws/install
Found preexisting AWS CLI installation: /usr/local/aws-cli/v2/current. Please rerun install script with --update flag.
ubuntu@ip-172-31-25-174:~$ aws --version
aws-cli/2.31.27 Python/3.13.9 Linux/6.14.0-1015-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-25-174:~$ aws s3 ls
Unable to locate credentials. You can configure credentials by running "aws configure".
ubuntu@ip-172-31-25-174:~$
```

**i-0d05f05503f102671 (serverlab4)**  
PublicIPs: 13.51.157.162 PrivateIPs: 172.31.25.174

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```
inflatng: aws/dist/wheel-0.45.1.dist-info/METADATA
inflatng: aws/dist/wheel-0.45.1.dist-info/RECORD
inflatng: aws/dist/wheel-0.45.1.dist-info/direct_url.json
inflatng: aws/dist/wheel-0.45.1.dist-info/LICENSE.txt
inflatng: aws/dist/wheel-0.45.1.dist-info/REQUESTED
inflatng: aws/dist/wheel-0.45.1.dist-info/entry_points.txt
ubuntu@ip-172-31-25-174:~$ unzip awscli2.zip
Archive:  awscli2.zip
replace aws/README.md? [yes, [no], [All], [None], [r]ename: ^X
error:  invalid response []
replace aws/README.md? [yes, [no], [All], [None], [r]ename: ^Cubuntu@ip-172-31-25-174:~$ n
n: command not found
ubuntu@ip-172-31-25-174:~$ sudo ./aws/install
You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-25-174:~$ sudo ./aws/install
Found preexisting AWS CLI installation: /usr/local/aws-cli/v2/current. Please rerun install script with --update flag.
ubuntu@ip-172-31-25-174:~$ aws --version
aws-cli/2.31.27 Python/3.13.9 Linux/6.14.0-1015-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-25-174:~$ aws s3 ls
Unable to locate credentials. You can configure credentials by running "aws configure".
ubuntu@ip-172-31-25-174:~$ aws configure
AWS Access Key ID [None]: AKIA4VHPU66NKG2VXORS
AWS Secret Access Key [None]: JmLOGM42d1No0NSx//ZB8rpoWVTvnl1wThod1m
Default region name [None]:
Default output format [None]:
ubuntu@ip-172-31-25-174:~$
```

**i-0d05f05503f102671 (serverlab4)**  
PublicIPs: 13.51.157.162 PrivateIPs: 172.31.25.174

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Now this runs

```
ubuntu@ip-172-31-25-174:~$ aws s3 ls
2025-11-01 08:35:26 rameen-s3-bucket-22k5167
ubuntu@ip-172-31-25-174:~$
```

Nano [main.tf](https://main.tf) and paste



The screenshot shows a web browser window with multiple tabs open, including 'my-notification', 'Instances | EC2 | eu-north', 'EC2 Instance Connect | eu', 'Upload objects - S3 bucket', 'snsfunction-role-v93voo', and 'AWS Policy Generator'. The active tab is 'eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0d05f05503f102671&osUser=ubuntu&region=...'. The browser's address bar shows the URL. The main content area displays a terminal window with the following Terraform configuration:

```
GNU nano 7.2 main.tf *
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 6.0"
    }
  }

  required_version = ">= 1.2.0"
}

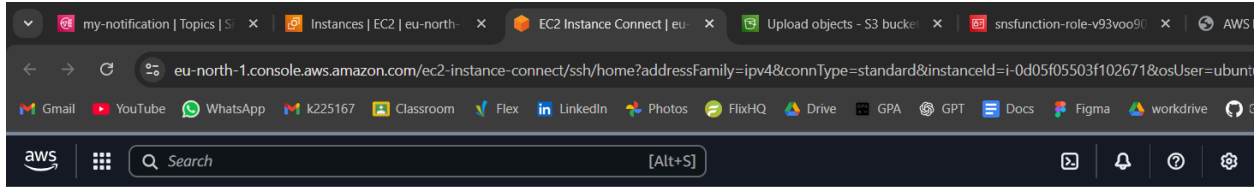
provider "aws" {
  region = "us-east-1"
}

resource "aws_instance" "my_ec2_instance" {
  ami          = "ami-08c40ec9ead489470" # Example Ubuntu AMI
  instance_type = "t2.micro"

  tags = {
    Name = "Instance-by-TF"
  }
}
```

Below the terminal window, the instance ID 'i-0d05f05503f102671 (serverlab4)' is displayed, along with its Public IP '13.51.157.162' and Private IP '172.31.25.174'. The bottom of the browser window shows the 'CloudShell' header with 'Feedback' and 'Console Mobile App' links, and the footer with '© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences'.

- Change t2 to t3
- Run terraform init
- Plan means details of everything to make
- Run terraform plan
- Known after apply - by default will be selected
- Run terraform apply



```
+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ primary_network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device (known after apply)
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.my_ec2_instance: Creating...
aws_instance.my_ec2_instance: Still creating... [00m10s elapsed]
aws_instance.my_ec2_instance: Creation complete after 16s [id=i-015303ef185d0dcb8]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
ubuntu@ip-172-31-25-174:~$
```

i-0d05f05503f102671 (serverlab4)  
PublicIPs: 13.51.157.162 PrivateIPs: 172.31.25.174

ec2 instance would have been created now  
Change your region to east 1 to view it

